

A PROJECT REPORT ON

BANK MANAGEMENT SYSTEM

Submitted to fulfillment of II Year I Semester Lab Exam

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

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CERTIFICATE

This is to certify that the project titled “**Bank Management System**” is being submitted by **P.MANASWINI(24911A67B5),S.NAVANEETH(24911A67A2),M.AKSHAYA (24911A67A2),L SAGAR(24911A6798),B BHAVYA SRI(24911A6772)**. In fulfillment of **B.Tech II Year I Semester Data Structures Lab Exam** in **Computer Science & Engineering (Data Science)**, is a record of Bonafide work carried out by them under my guidance and supervision. These results embodied in this project report have not been submitted to any other University or Institute for any purpose.

Internal Guide
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DECLARATION

We, **P.MANASWINI, S.NAVANEETH, M.AKSHAYA, B BHAVYA SRI , L SAGAR** bearing Roll Numbers **24911A67B5, 24911A67B9, 24911A67A2, 24911A6798, 24911A6772** hereby declare that the project entitled, “**Bank Management System**” submitted for the fulfillment of **B.Tech II Year I Semester Data Structures Lab Exam** in **Computer Science & Engineering (Data Science)**, is original and has been done by us and this work is not copied and not submitted anywhere for any purpose.

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Place: Hyderabad

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INTRODUCTION

The *Bank Management System* is a console-based application developed using the C programming language and fundamental *Data Structures. The project simulates basic banking operations such as creating accounts, depositing and withdrawing money, viewing account details, and deleting accounts. The main objective is to demonstrate the practical use of data structures like **structures, arrays, linked lists, and files* in real-world applications.

Objectives

The main objectives of the project are:

- To understand and apply data structures in C
- To manage bank account records efficiently
- To perform basic banking transactions
- To implement file handling for data storage
- To provide a menu-driven user interface

Scope of the Project

This project is designed for academic purposes and small-scale usage. It does not include advanced security, online transactions, or real-time banking features. The system supports:

- Account creation
- Deposit and withdrawal
- Balance enquiry
- Account deletion
- Viewing all accounts

System Requirements:

Hardware Requirements

- Processor: Intel i3 or above
- RAM: Minimum 4 GB
- Hard Disk: 100 GB
- Keyboard and Mouse

Software Requirements

- Operating System: Windows / Linux
- Programming Language: C
- Compiler: GCC / Turbo C
- Code Editor: VS Code / Dev-C++

Overall Design

Data Structures Used

Structure

A structure is used to store bank account details:

- Account Number
- Account Holder Name
- Account Type
- Balance

Linked List

- Each node represents a bank account
- Allows dynamic memory allocation
- Efficient insertion and deletion of records

Modules Description

Create Account

- Accepts customer details
- Generates a unique account number
- Stores data in linked list and file

Deposit Amount

- Allows adding money to an account
- Updates balance

Withdraw Amount

- Allows withdrawing money
- Checks minimum balance condition

Balance Enquiry

- Displays current balance of an account

Display All Accounts

- Shows details of all customers

Delete Account

- Removes account from linked list
- Updates stored data

Flow of Control (Algorithm)

Algorithm (High-level):

Step 1:Start the program

Step 2:Display menu options

Step 3:Accept user choice

Step 4:Perform selected operation

Step 5:Repeat until exit option is selected

Step 6:Stop the programYou can also draw a flowchart based on this algorithm for your report.

Implementation Details

File handling

- Used to store account details permanently
- Files are read when program starts
- Files are updated after each transaction

Advantages

- Simple and user-friendly
- Efficient data management
- Demonstrates core data structure concepts

Limitations

- No graphical user interface
- No encryption or security
- Single-user system

Future Enhancements

- Add login authentication
- Implement database (MySQL)
- Add GUI using C graphics or other languages
- * Support online transactions

CONCLUSION

The Bank Management System developed using C and data structures provides a clear understanding of how banking operations can be implemented programmatically. This project helps students gain practical experience in data structures, file handling, and modular programming.

OUTPUT

```
-----  
      BANK MANAGEMENT SYSTEM  
-----  
1. Create New Account  
2. Deposit Amount  
3. Withdraw Amount  
4. Balance Enquiry  
5. Display Account Details  
6. Exit  
Enter your choice: 1  
Enter Account Number: 1001  
Enter Account Holder Name: Manaswini  
Enter Initial Deposit Amount: 5000  
Account created successfully!  
Enter your choice: 2  
Enter Account Number: 1001  
Enter Deposit Amount: 2000  
Amount deposited successfully!  
Updated Balance: 7000  
Enter your choice: 3  
Enter Account Number: 1001  
Enter Withdrawal Amount: 151500  
Withdrawal successful!  
Remaining Balance: 5500  
Enter your choice: 4  
Enter Account Number: 1001  
Current Balance: 5500  
Enter your choice: 5  
Account Number:      1001  
Account Holder Name: Manaswini  
Account Balance: 5500  
Thank you for using Ban Management System!
```